

Antonio Carmagnani

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Dear Hiring Team at NTNU,

I trained as an architect, and the interest that has run alongside that from the beginning is data - using it to understand what has been built and what could be built better. That is what took me from architecture practice into feasibility and analysis, and then to a Master's Degree in Architecture Computation at The Royal Danish Academy, a programme built around using data and technology to improve the decisions made in real estate and construction. The through line has been consistent, which is that I would rather build the analytical tool that answers a question than argue the question without one. NTNU asks the harder version of that question, how climate risk and intersectional vulnerability are read together when one is far better evidenced.

My Master's Degree in Architecture Computation gave the analytical side of my work its proper depth. I built a predictive property valuation model using ensemble learning across XGBoost, LightGBM, Ridge and CatBoost that finished in the top two per cent of its competition, and a machine learning tool that generates site-specific development recommendations for buildings in Copenhagen by combining financial, social, sustainability and geospatial data, with an interface that lets non-technical stakeholders adjust the impact drivers and weightings themselves. The second of those mattered more to me than the first, because it forced the modelling to survive contact with people who would never open the code. The project asks for that integration directly, spatial and non-spatial data brought into vulnerability maps that stakeholders can contest.

At Sweco I supported advisory teams delivering water, energy and transport projects financed by development banks including the World Bank, EBRD, KfW and EIB, and authored climate risk assessments for major water supply projects under JASPERS and EBRD climate-proofing guidance, screening hazards against projection data and rating exposure and vulnerability asset by asset. Working to lender standards taught me that an assessment is only useful if it survives review by someone with money at stake, which changes how you evidence every claim you make. Validating AI-generated scenarios against established planning tools rests on that same discipline of evidence.

CREST-GLOBAL works with Norwegian municipalities on equitable urban transitions, keeping the research answerable to the places it describes. That is where I want to keep building sustainability work that is measurable, backed by evidence, and able to withstand scrutiny.

Best,

Antonio Carmagnani